Recovering Code Performance After Persona Fine-Tuning: A Single-Run Case Study on Qwen2.5-3B-Instruct

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CS 2881R, Assignment 1

github.com/essacheez/Posttraining-coding-rocky

Abstract

We fine-tune Qwen2.5-3B-Instruct to answer in the voice of Rocky from *Project Hail Mary* while retaining Python code generation. Code accuracy is reported as pass@1, the fraction of problems solved by the model's first generated program. Supervised fine-tuning (SFT) changes surface style but reduces HumanEval and HumanEval+ pass@1 by 20.7 and 23.2 percentage points. An initial reinforcement learning from AI feedback (RLAIF) run exploits an underspecified style reward; a revised objective separates task, style, format, and anti-catchphrase rewards. A final RLVR stage trained on a 700-problem, difficulty-balanced curriculum recovers 11.6 points on HumanEval, 10.9 on HumanEval+, and 2.2 on MBPP in the training-trajectory evaluation. A separate final-model rerun obtains 0.683, 0.628, and 0.584, respectively, plus 0.149 on LiveCodeBench. The completed five-dimension persona judge scores the final model 18.5/25, including 3.19/5 for human plausibility. Each training stage was run once, so these results do not measure variation across repeated training runs. The evidence shows substantial recovery of the measured coding loss while retaining recognizable persona features, but not the absence of a general capability–persona trade-off.

1 Introduction

Post-training can install a desired behavior while degrading capabilities that the training data rarely exercises. Persona tuning makes this failure easy to observe: language style and task content share the same output channel, so examples that reward stylized prose can implicitly discourage code. We test this interaction using Qwen2.5-3B-Instruct [8], a low-rank adaptation (LoRA) pipeline [4], and a target persona with conspicuous grammatical markers.

The project asks three empirical questions. First, how much coding ability is lost when the persona is installed by SFT? Second, can an AI judge improve the persona without being gamed? Third, can execution-based reinforcement learning recover coding ability without undoing the persona? We use SFT, direct RLAIF [2, 5], and group relative policy optimization (GRPO) [7] in sequence.

Our contributions are primarily diagnostic:

- We document a data-composition failure: only 21 of 1,000 SFT targets contain a fenced code block.
- We retain a failed RLAIF checkpoint as a control and show that a holistic persona score can miss obvious reward gaming.
- We compare earlier MBPP-based RLVR ablations with a later run described in its metadata as a 700-problem benchmark-disjoint curriculum, while documenting that the exact curriculum and builder are missing from the committed repository.
- We report the completed coding, verifier, stylometric, and five-dimension persona evaluations, including disagreements between metrics and reruns.

2 Experimental Setting

2.1 Model and training sequence

All stages adapt Qwen2.5-3B-Instruct using rank-32 LoRA modules on the model's projection layers. The deployed sequence is:

1. **SFT:** 1,000 rewritten assistant answers are split 95/5, leaving 950 training rows and 50 validation rows. Training uses three epochs and response-only loss.
2. **RLAIF:** the builder assembles 600 rows: 500 assistant prompts, 60 narrative prompts drawn from the SFT training file, and 40 MBPP validation problems. A 95/5 split leaves 570 rows for one epoch. Eight generations are sampled per prompt and optimized with GRPO. GPT-5.4-nano supplies task and style rewards, while deterministic rules score formatting and catchphrase frequency.
3. **Curriculum RLVR:** run metadata and the plotting script describe 700 code problems selected from a 1,700-problem pool, trained for 174 steps over two epochs on two L40S GPUs. The exact dataset is absent from the branch; the tracked `rlvr_dataset.jsonl` is an older 677-row MBPP/persona mixture with a different schema.

SFT and RLAIF use the default Qwen chat template and no persona system prompt at evaluation time. The intended behavior must therefore be carried by the adapter rather than injected in the prompt.

2.2 Evaluation

We report greedy-decoding pass@1—the fraction of tasks solved by the model's first generated program—on Hu-

manEval (164 problems) [3], HumanEval+ (the EvalPlus augmentation of HumanEval) [6], and the 500-problem MBPP test split [1]. The latest run is described as excluding these benchmark problems, but the missing curriculum prevents repository-level verification. HumanEval+ reuses the 164 HumanEval prompts with substantially expanded tests; it is a stricter test suite, not an independent third problem set. All three reported test suites are evaluated at multiple checkpoints, so they function as development benchmarks for checkpoint inspection rather than untouched final tests.

Persona evaluation uses 200 held-out prompts. The primary automatic metric is cosine similarity between function-word, punctuation, and rhythm feature vectors and the target corpus. We also record centroid distance. These handcrafted metrics are not validated measures of character fidelity. The reference answers are model-written rewrites of human answers, not quotations from the novel, so they measure fidelity to our training target rather than fidelity to canon.

Table 1 reports the training-trajectory artifacts. A separate final-model rerun appears in Table 3. We do not average the two snapshots. Because each training stage was run only once and paired confidence intervals are not yet computed, stage-to-stage differences are descriptive rather than estimates of run-to-run variation.

3 SFT Installs Style and Damages Coding

The SFT set is built by rewriting 1,200 human-written conversations from No Robots, Dolly-15k, and OASST2; 1,000 enter the SFT dataset, 950 are used for optimization after the internal split, and 200 separate rows are used for persona evaluation. This construction holds the question and intended content approximately fixed while changing the response style. It does not hold answer quality exactly fixed because the rewrite model can alter content.

SFT raises style cosine from 0.583 to 0.879, but HumanEval falls from 0.762 to 0.555 and HumanEval+ from 0.738 to 0.506. The data composition offers a plausible mechanism: only 21 of the 1,000 SFT rows use a fenced code block, and the rewritten answers are terse. In one evaluation artifact all 164 base completions contain a code block while none of the 164 SFT completions do. That observation supports a formatting-loss hypothesis, but the repository does not yet contain a paired decomposition of formatting errors versus incorrect program logic. We therefore avoid attributing the entire coding decline to formatting.

An earlier synchronized ablation is important context. SFT with self-generated code targets reaches 0.701 on HumanEval and 0.689 on HumanEval+, with an aggregate persona-judge score of 19.725. In that evaluation snapshot it outperforms the earlier multi-objective RLVR checkpoint (0.622 and 0.573; judge 19.271). Those values are not directly comparable to the later curriculum run because the evaluation snapshots differ, but they show that a data intervention is a serious alternative to adding RL stages.

4 RLAIF and a Reward-Gaming Control

4.1 Failed first attempt

The first RLAIF run uses a single judge reward and a prompt that explicitly names catchphrases as desirable. Its held-out reward rises while coding collapses; by the end, 91% of persona replies open with "Listen. Listen." This is a useful control because the outputs visibly optimize a recognizable proxy instead of the intended natural voice. The absolute size of the code collapse is uncertain: the historical evaluator extracted only the first code block, while the model often split code across blocks.

4.2 Revised objective

The revised run separates four signals (Table 2) and increases the KL coefficient from the failed configuration. The decomposition makes it possible to see which proxy is changing rather than hiding all behavior in one scalar reward.

RLAIF suppresses the most conspicuous learned tics but does not demonstrate improved overall character fidelity. Corpus-level style cosine rises from 0.879 to 0.930, although a prompt-level bootstrap interval for the change includes zero (-0.02 to +0.13). In the blind canon-grounded sample, RLAIF minus SFT is -0.075 points on a five-point scale (95% interval -0.25 to +0.10). The clearest observed change is that forced-catchphrase flags fall from 42% to 0%. The penalty also fires on 58% of synthetic target-style references, indicating that it is stricter than the distribution it is meant to preserve.

5 Verifiable-Reward RL

5.1 Earlier MBPP-based ablations

The earlier `rlvr` branch contains verifier-only and multi-objective experiments trained on MBPP-derived problems. They test whether executable reward helps and whether persona rewards should remain active during RLVR. However, the training prompt exposes the same asserts used by the verifier, and MBPP is also an evaluation benchmark. Those runs are informative ablations but weak evidence of held-out generalization. We preserve their code and results while excluding them from the headline causal claim.

5.2 Difficulty-balanced curriculum

The final RLVR stage uses a curriculum: training problems are grouped and ordered by estimated difficulty rather than sampled as one undifferentiated set. The run description says that this stage begins from the RLAIF adapter and excludes the three reported test suites. Candidate difficulty is estimated from eight sampled RLAIF completions per problem. The selected set contains every easy and medium problem, 316 of 318 hard problems, no already-solved problems, and 70 of 1,030 problems for which all eight samples fail. This concentrates

Table 1: Training-trajectory checkpoint results. Coding entries are pass@1. "Style cosine" is similarity to the project target corpus.

Checkpoint	HumanEval	HumanEval+	MBPP	Style cosine
Base Qwen	**0.762**	**0.738**	**0.588**	0.583
SFT	0.555	0.506	0.556	0.879
RLAIF	0.579	0.537	0.562	0.930
Curriculum RLVR, step 174	0.695	0.646	0.584	**0.947**

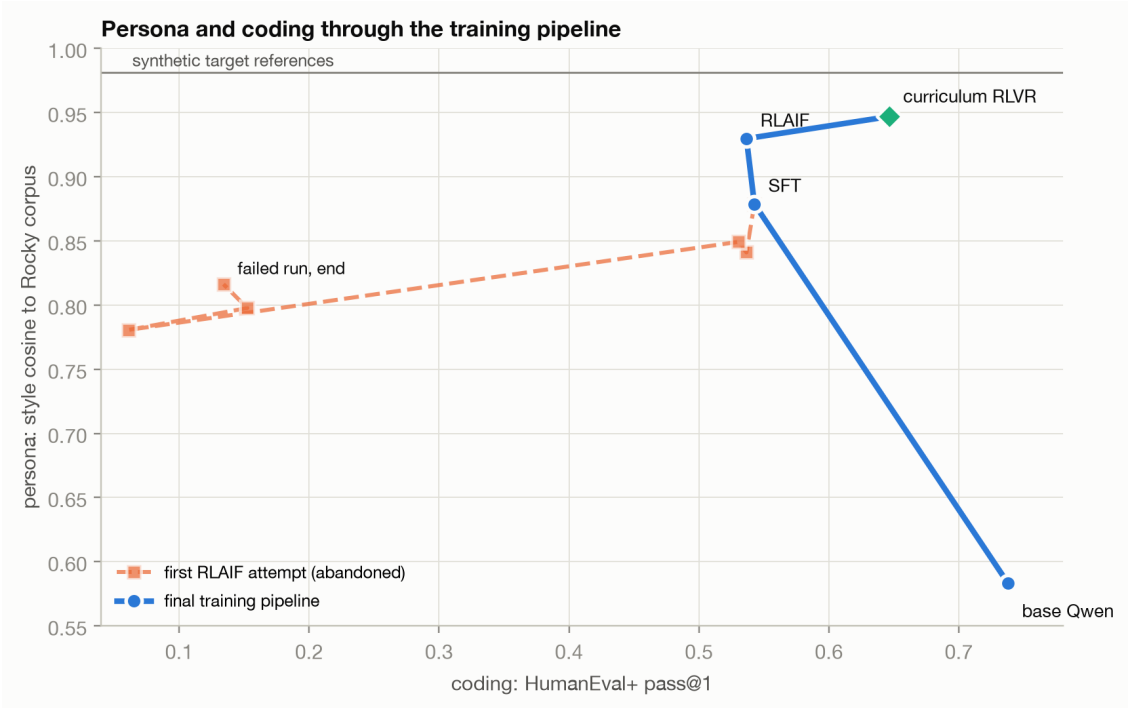


Figure 1: Coding and persona style through the main training path, with the abandoned first RLAIF run retained as a failure control. The apparent improvement in both axes at the curriculum checkpoint is subject to the metric and replication limitations in Sections 6 and 7.

Table 2: Reward terms in the revised RLAIF run.

Term	Measurement	Weight
Task	AI judge: answer correctness	1.0
Style	AI judge: target voice	0.6
Catchphrase	Rule-based excess-frequency penalty	1.0
Format	One code block; response-length guard	1.0

Table 3: Two evaluations of the final RLVR checkpoint. A dash means the benchmark was not included. Values are reported separately, not averaged.

Evaluation	HumanEval	HumanEval+	MBPP	LCB
Training trajectory	0.695	0.646	0.584	–
Shared final rerun	0.683	0.628	0.584	0.149

training on problems that are neither trivial nor uniformly uninformative, while retaining a small very-hard slice. These counts are hard-coded in the plotting script. The 700 rows and their builder are not committed, so the composition and deduplication claim cannot currently be audited.

The binary reward requires exactly one fenced Python block and all visible tests to pass. It supplies no persona or AI-judge reward. In the recorded evaluations, curriculum RLVR moves from 95/164 to 114/164 passing HumanEval completions, 88/164 to 106/164 on HumanEval+, and 281/500 to 292/500 on MBPP relative to RLAIF (Figure 5). The last checkpoint remains below the recorded base by 11 HumanEval problems, 15 HumanEval+ tests, and two MBPP problems. These comparisons use one training run and mix evaluation launches; they are development-set evidence, not an unbiased estimate of final generalization.

The shared final-model rerun produces slightly different HumanEval and HumanEval+ generations from the training-trajectory evaluation (Table 3). Greedy vLLM generation was not bitwise repeatable in our systems check, so we report both snapshots rather than choosing the larger one. The matching MBPP result and separate rule-based verifier provide a partial consistency check.

Qualitative inspection of disagreements with the base model suggests many residual errors are semantic near misses: short programs solve a simpler interpretation of the prompt. This is a hypothesis generated from examples, not a coded error analysis. A blinded taxonomy with inter-rater agreement would be

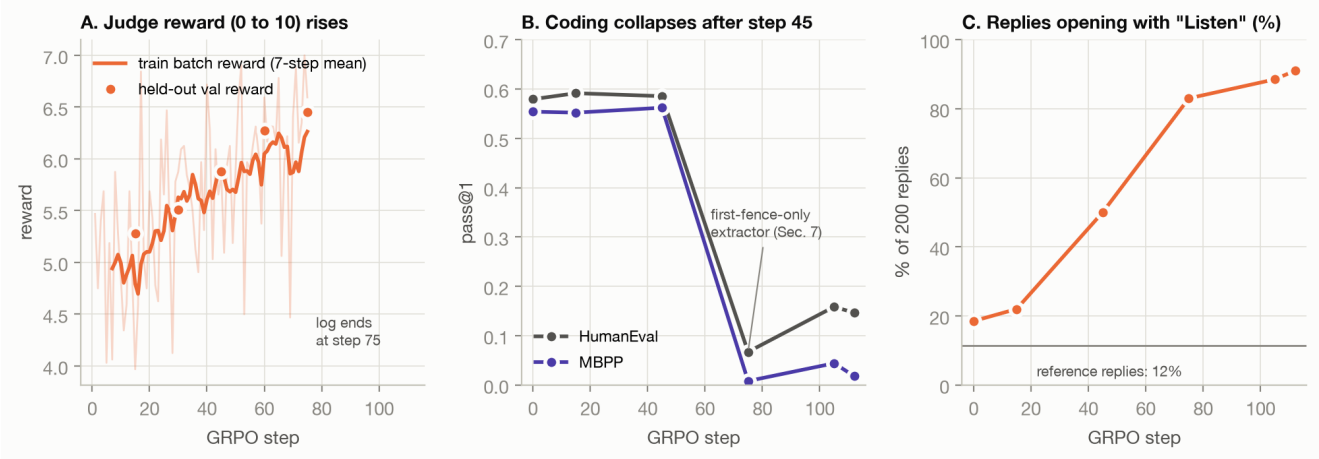


Figure 2: The abandoned first RLAIF run. Judge reward rises, the historical coding measurement collapses after step 45, and one rewarded opener saturates the outputs. The coding curve is partially confounded by the old first-block extractor.

needed to make it a result.

6 Does the Persona Survive?

Style cosine increases from 0.930 after RLAIF to 0.947 after curriculum RLVR. This is a corpus-level cosine over aggregated style features. Per-response centroid distance, however, worsens from 1.147 to 1.208. The aggregation difference can produce opposite movements, so the disagreement prevents a simple claim that persona quality improved. The most defensible current statement is that the two non-judge metrics are inconclusive.

We also conducted two blind reads on earlier checkpoints. One teammate rated 16 replies. Two Claude model-rater instances scored 160 shuffled replies (40 prompts by four sources) against a separately researched canon profile. The target rewrites beat SFT by 0.35 points on a five-point fidelity scale (95% interval 0.08 to 0.63; Wilcoxon $p = 0.02$). This is one of several exploratory comparisons and would not remain below 0.05 under a simple Bonferroni correction across the three displayed tests. RLAIF minus SFT is -0.08 (interval -0.25 to +0.10). The known-gamed first RLAIF run also scores about the same as SFT on holistic fidelity, while a targeted flag identifies forced catchphrases in 98% of its replies. Thus the holistic rater fails an important positive-control test.

This evaluation has substantial limitations. The canon rater is itself a model, each reply is scored by only one of two rater instances, the human sample has four replies per source, and only non-code prompts are rated. More fundamentally, the rewrite prompt teaches habits that are weakly grounded in the novel: for example, "Listen. Listen." is frequent in the training target, while the book-derived profile does not support it. All automatic style scores can therefore reward fidelity to a synthetic target that is not faithful to the character.

The completed GPT-5.4 rubric evaluation scores all 200 final-model replies: sentence structure 4.675, vocabulary 4.280, markers 2.770, consistency 3.585, and human plausibility 3.190, for 18.5/25 overall. Its markers-minus-plausibility gap is -0.42, which does not show the marker-heavy pattern seen in the failed RLAIF control. The rubric uses the same project style construct that informed the training target, however, so it is corroborating evidence rather than independent human ground truth.

7 Limitations and Threats to Validity

No replicated training runs. Each training stage was run once. We therefore cannot measure how results vary across random seeds or separate method effects from sampling and systems variance. Even greedy vLLM generations were not bitwise repeatable in a repository check.

Checkpoint selection and repeated evaluation. The latest curriculum checkpoint is reported after inspecting benchmark trajectories. Calling a point "best" is descriptive; it is not an unbiased held-out selection. A clean confirmation would choose a checkpoint using training reward or a validation set and evaluate the three test suites once.

Incomplete uncertainty. For 164-problem benchmarks, a single pass rate has a rough binomial uncertainty of several percentage points. Stage comparisons are paired by problem, so a paired bootstrap or McNemar analysis is preferable to independent binomial intervals. Those paired results are not yet included.

Metric construction. Style cosine, centroid distance, catchphrase flags, and AI rubrics were all designed by the project team. They are neither independent nor externally validated, and some target the same markers used during training. The completed rubric judge is also an AI system, not human ground truth.

Missing curriculum and data contamination. The run is described as excluding the named evaluation problems, but the

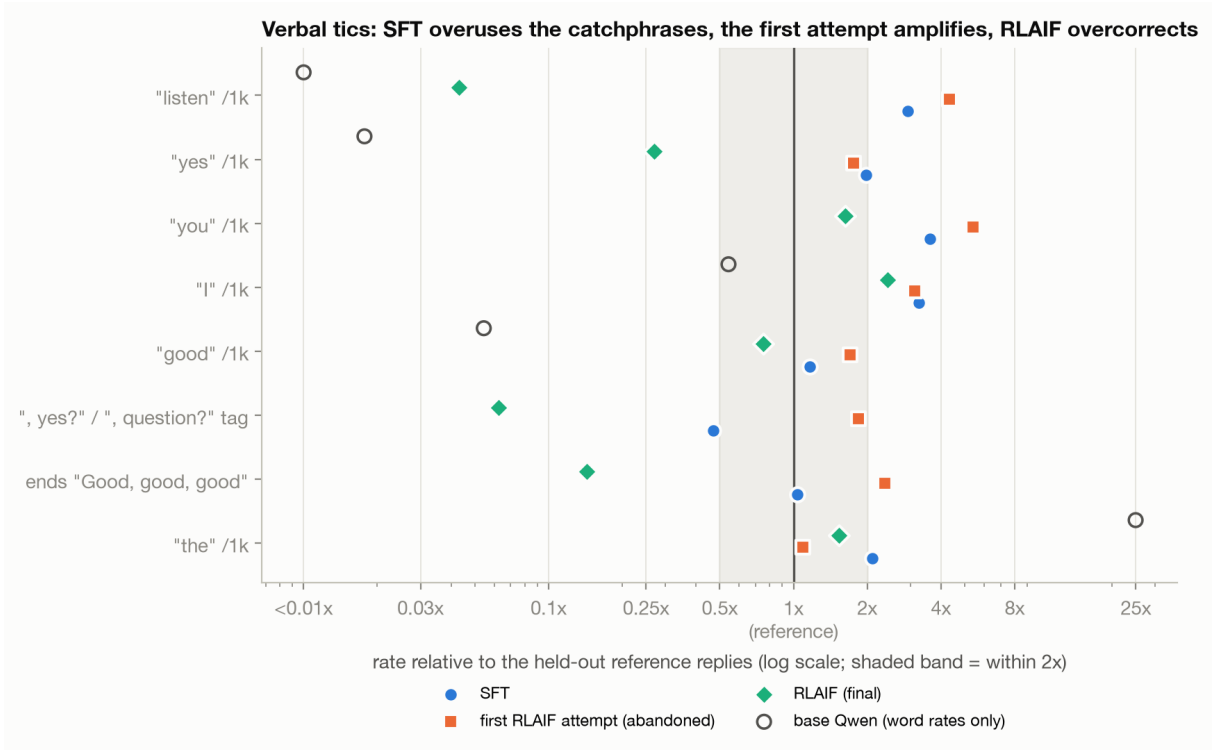


Figure 3: Rates of selected verbal markers relative to held-out target rewrites. The revised RLAIF run suppresses the targeted catchphrases, including some that occur in the target itself. The figure measures feature frequency, not human-likeness or canonical character fidelity.

exact 700-row dataset and builder are absent. The committed 677-row file belongs to an older MBPP/persona experiment and does not satisfy the current trainer's `test_cases` schema. We therefore cannot audit direct overlap, near duplicates across public code corpora, or the base model's pretraining data.

Changing more than one variable. The failed and revised RLAIF runs differ in prompt, reward decomposition, learning rate, and KL coefficient. Their contrast diagnoses a bad configuration but does not isolate which change fixes it.

8 Conclusion

The experiment begins with a common post-training failure: SFT successfully installs a surface persona while degrading a capability absent from most demonstrations. A naive AI-judge objective then creates an obvious proxy exploit. Reward decomposition removes that specific observed pattern, and the reported executable-reward curriculum recovers much of the measured coding loss. The result does not establish that persona and coding are generally compatible, nor that the latest checkpoint is preferable on every persona measure. It does show why failure controls, disaggregated rewards, auditable data, a separate final test, and multiple style metrics are necessary for making a credible claim.

Reproducibility Statement

Training code, evaluation outputs, adapters, figure scripts, and this LaTeX source are versioned in the project repository. All scheduled evaluations are complete, including the final persona judge, verifier check, and LiveCodeBench. Their outputs are under `EVALS/results_rlvr/extra_checks`. Training reproduction remains incomplete: the latest 700-row curriculum and its builder are missing, while the committed RLVR dataset is from an older experiment and has an incompatible schema. The main result files are `EVALS/results_base`, `EVALS/results_rlaif`, and the top-level files in `EVALS/results_rlvr`. Historical MBPP-based ablations are stored in named subdirectories under `EVALS/results_rlvr`. The paper figures are regenerated by `report/scripts/fig_*.py` and `RLVR/figures/plot_rlvr.py`. The declared Python environment also needs repair: its pinned Transformers version is incompatible with the pinned sentence-transformers requirement. We therefore do not claim one-command reproduction until the dataset and environment issues are fixed.

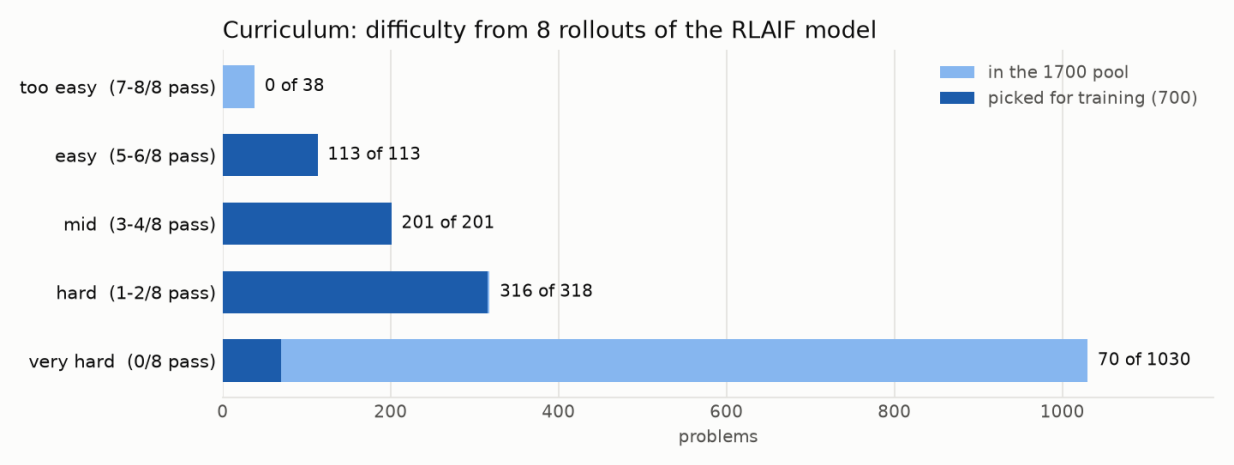


Figure 4: Reported curriculum construction from a 1,700-problem pool. Difficulty is the RLAIF model's success count across eight samples. Dark bars indicate the 700 reported selected tasks. The source rows are not committed; this figure reproduces counts encoded in the plotting script rather than recomputing them from data.

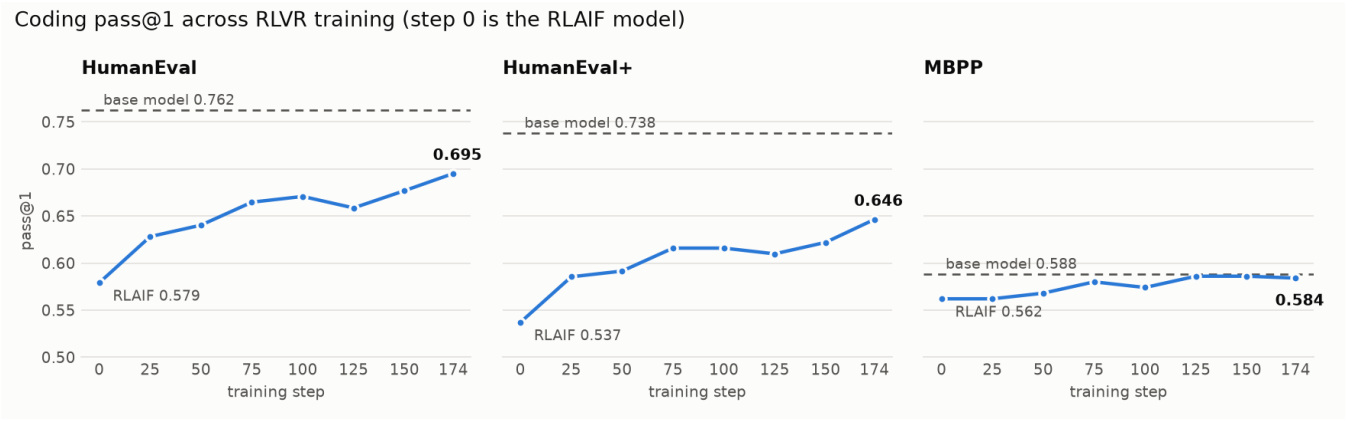


Figure 5: Coding pass@1 during curriculum RLVR. Step 0 is the RLAIF model. The final checkpoint is the best observed checkpoint on HumanEval and HumanEval+; MBPP peaks at 0.586 at steps 125 and 150, versus 0.584 at step 174. Repeated inspection makes all checkpoint comparisons exploratory.

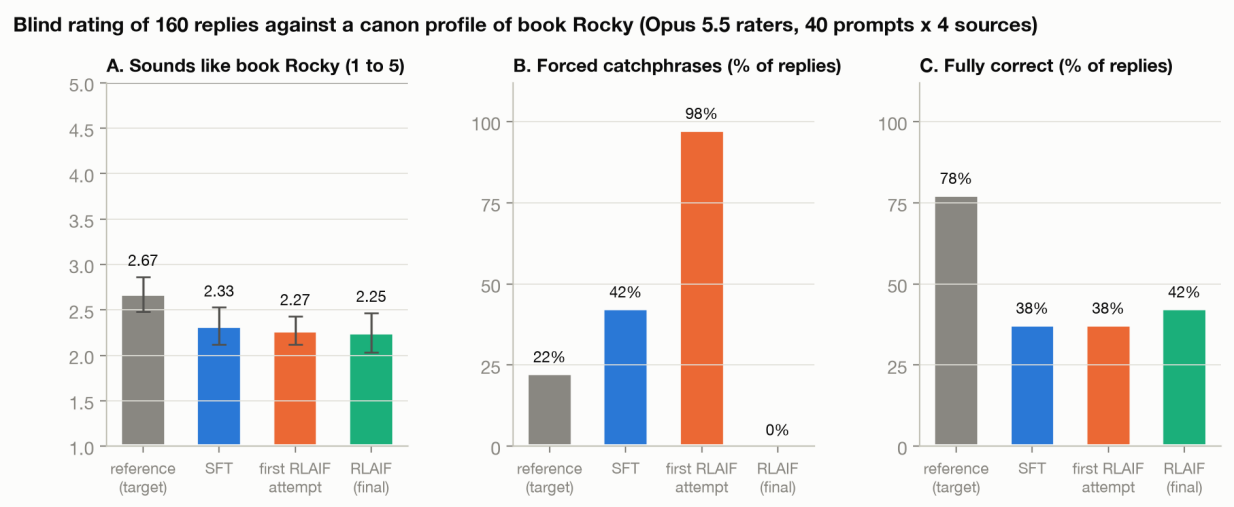


Figure 6: Blind canon-grounded evaluation of 160 replies from four earlier sources. The holistic 1-to-5 score does not separate the visibly gamed first RLAIF run from SFT; the targeted forced-catchphrase flag does. Error bars in panel A are 95% intervals.

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